Prediction of listened music features from EEG signal

(Predykcja cech słuchanej muzyki na podstawie sygnału EEG)

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Abstract

We consider the possibility of decoding music features from the EEG signal. We utilize datasets of joint music and EEG recordings. First, we ask whether the song name can be predicted [and get a mixed answer]. We explain the result with a DTW experiment. We further ask whether music features can be predicted. For that, we classify EEG snippets into "slow" and "fast" based on the tempo of the related music snippet.

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Chapter 1

Introduction

The EEG signal has been used in both research and clinical applications to classify brain states and diagnose various conditions, such as epilepsy or sleep disorders[**?**]. What initially required manual inspection is being superseded by automated computer methods. The ability to infer signals from EEG recordings has sparked hope for the eventual use of EEG in Brain-Computer Interfaces (BCI). Our work explores one such avenue.

We analyze datasets of EEG recordings while participants listened to music. We attempt to infer music properties from the EEG signals with machine learning methods. We succeed at the song classification task but observe poor cross-subject generalization. We try to explain this result with an experiment that attempts to perform song identification using a classical DTW algorithm.

We attempt emotion classification, which, despite being similar to song identification, shows poor results. We also define a novel classification task based on the number of note onsets in a fragment of a musical recording and solve it using our methods.

Lastly, we tackle the harder than classification problem of music decoding from EEG and achieve results good enough to perform classification on, but the poor generalization problem persists.

Chapter 2

EEG Signal

2.1 Overview

Electroencephalography (EEG) is a non-invasive technique used to record electrical activity of the brain[27]. It involves placing electrodes on the scalp to record electrical activity of the brain — the electroencephalogram (also EEG). Each electrode records the voltage fluctuations resulting from the synchronized discharges of the working neurons in the brain. By subtracting the reference electrode reading from a reading of an electrode of interest we cancel out the common environmental noise and get a signal that constitutes a single channel of an EEG recording. The amplitude of the resulting signal is in the order of microvolts and very sensitive to any head movements, which need to be minimized during a recording session. Modern equipment captures these fluctuations with upwards of 1000 Hz temporal resolution, allowing for precise linking between the brain's recorded activity and external stimuli.

2.2 History

A British physician, Richard Caton, was the first to discover in 1875 that the brain — specifically of rabbits and monkeys — elicits electrical activity and, furthermore, that this activity seems related to its function[4]. For his discovery, he used a galvanometer, invented earlier in 1820 and used in laboratories and telecommunication. At the time of his discovery, it was already known that nerves conduct electrical signals and that these signals are used to trigger muscle tissue to contract[23][24]. In 1901, a Dutch medical doctor, Willem Einthoven, invented a more precise string galvanometer and used it to record an electrocardiogram (ECG)[34]. By placing electrodes on the left and right arms and a leg, he measured the electrical signal produced by a beating heart. The device functioned by placing a fine conducting string within a magnetic field created by two powerful electromagnets; the elec-

trical signal caused this string to deflect. He was later awarded the Nobel Prize in Physiology or Medicine for his discovery. Later, in 1912, a Ukrainian physiologist, Vladimir Pravdich-Neminsky, used this invention and published the first EEG recording recorded from a dog[42].

In 1924 — which coincidently was the year when Einthoven was awarded his Nobel Prize — a German psychiatrist, Hans Berger, while looking for the physiological basis for telepathy, recorded the first human EEG and discovered the "alpha wave"—the frequency present in the signal when a person is awake[43]. His discovery was initially met with skepticism[27], which changed a decade later when Frederick Gibbs and others used EEG to describe specific, repeatable patterns in electrical activity during epilepsy[44]. Later, in the 50s, the EEG signal was used to research sleep. In 1953, researchers discovered the REM (Rapid Eye Movement) sleep phase, characterized by rapid eye movements and high-frequency, low-voltage EEG activity, akin to alertness, which is often the phase when dreaming occurs[1]. Later, a standardized way to divide sleep into sleep stages based on EEG was developed[16].

Already in 1973, the first demonstration of a brain-computer interface (BCI) was seen, where the EEG signal was used to control a cursor on a screen[45]. This decade also saw the first use of computers to analyze the EEG signal, and later the analysis and collection of EEG data became purely digital. In the 1980s, laboratories started syncing EEG recordings with video recordings of the subject, allowing for more detailed studies of the relationship between brain activity and behavior[14].

Modern-day 128- or 256-electrode systems allow for spatially detailed recordings of brain activity, including source localization. EEG is now widely used in clinical settings for the diagnosis of sleep disorders[18] and the localization and classification of seizures[11]. We also see a rise in the availability of consumer-grade portable EEG devices, which are used for various applications, including neurofeedback, meditation, and sleep improvements.

2.3 Characteristics

The EEG signal has a few characteristics that make it extra challenging to process and analyze.

2.3.1 Non-stationarity

A non-stationary process is one whose statistical properties — like mean, variance, or autocorrelation — change over time. The EEG signal is non-stationary[25] because the brain's activity is dynamic and influenced by various internal and external factors, such as cognitive processes, sensory inputs, emotional states, and physiological changes like variations in skin impedance. Without any adaptations, this

renders many classical statistical methods of data analysis ineffective. This suggests that the normalization of signal data should be done on a per-window basis.

2.3.2 Artifacts

Because the amplitude of neural oscillations as seen in the EEG signal is so low, requiring precise measurement, even small movements and external factors can introduce significant noise and artifacts into the recordings. Specifically, any head movements or eye blinks cause large artifacts in the signal. It can also happen that an electrode loses contact with the skin or that heartbeats appear from an electrode placed near a large blood vessel. Given the small sample size, this often necessitates marking and removing the artifacts from the data.

2.3.3 Expensive and time-consuming data collection

Collecting EEG data requires specialized, expensive equipment and qualified personnel. It also requires finding subjects willing to participate in the study and fitting the study requirements. For these reasons, EEG datasets are often small, counting hundreds to thousands of samples collected from up to a few dozen subjects.

2.3.4 Noise

Given the low amplitude of the EEG signal, it is very susceptible to any external noise. Commonly, EEG data preprocessing includes filtering the 50 or 60 Hz power line noise, which appears as thick fuzz on the signal. In studies that combine music listening with EEG collection, the music itself would be a source of noise. To mitigate this, studies can use electromagnetically shielded insert earphones[28], which pump the sound directly into the ear via hollow plastic tubes, keeping the transducer (the part that creates sound) further away from the EEG apparatus.

2.3.5 Signal-to-noise ratio

Given that EEG measures a combined electrical activity of millions of neurons from a brain that reacts to and processes a multitude of internal and external stimuli, it is a given that the observed reaction to the experimental stimulus as witnessed from the data is going to be mixed with a lot of other brain activity, which is not related to the stimulus. Furthermore, the observed reaction is going to be different across subjects and even across different recording sessions of the same subject. For this reason, the signal-to-noise ratio (SNR) of the EEG data is often very low, making it difficult to extract meaningful information from the data and to link it to the experimental stimulus. A common way of analyzing EEG data involves averaging the EEG response at fixed intervals after presenting the stimulus.

2.4 Preprocessing

To mitigate the above challenges, various preprocessing techniques are used to clean the data and extract relevant features. We generally consider the EEG signal to be a two-dimensional array of shape (n_channels, n_timepoints), where n_channels is the number of electrodes used in the recording and n_timepoints is the number of time points recorded for each channel.

2.4.1 Filtering

Filtering is a crucial preprocessing step in EEG signal processing. It involves removing unwanted frequency components from the signal, which can be caused by various sources of noise and artifacts. Commonly, the signal is processed through a high-pass filter (e.g., 1Hz) to remove slow drifts caused by factors like sweating and changes in skin impedance. It can also be processed through a low-pass filter (e.g., 30Hz-100Hz) to remove high-frequency noise and focus on much better-understood lower frequencies in the EEG data. The power noise is removed with a notch filter at 50 or 60 Hz, depending on the local power line frequency of the recording environment.

Band power processing

A common signal processing technique — also applied to EEG data — is to filter the signal into chosen frequency bands and then calculate the power of the signal in each band, which is a measure of the strength of the oscillations in that band. Over the years, an array of research has been done linking specific frequency bands in the EEG signal to specific cognitive processes, with different roles linked to different frequency bands[2]. This guides the choice of frequency bands to be used for band power processing.

Band	**Frequency range**	**Associated states / processes**
Delta	0.5–4 Hz	Deep sleep, unconscious processes[15]
Theta	4–8 Hz	Drowsiness[20], meditation[19], memory encoding[17]
Alpha	8–13 Hz	Relaxed wakefulness, eyes closed, idle state[5]
Beta	13–30 Hz	Active thinking, concentration, alertness[10][33]
Gamma	30–100 Hz	Higher cognitive functions[41][13]

Table 2.1: EEG frequency bands and their associated cognitive states.

2.4.2 Independent Component Analysis

Independent Component Analysis (ICA) is a method of separating recorded EEG channels into independent components. Concretely, naming the EEG signal $\mathbf{x}$ and the result of applying ICA $\mathbf{s}$, ICA finds an inverse linear transformation to a matrix $\mathbf{A}$ and some $\mathbf{s}$:

$$\mathbf{x} = \mathbf{A}\mathbf{s}$$

such that the channels of $\mathbf{s}$ are:

- statistically independent
- non-Gaussian
- linearly mixed in the observed signal $\mathbf{x}$

Here $\mathbf{x}$ is of dimension (n_channels, n_timepoints) and $\mathbf{s}$ is of dimension (n_components, n_timepoints), where n_components is chosen and $\mathbf{A}$ is of dimension (n_channels, n_components).

In this formulation, the solution $\mathbf{s}$ is not unique, and specifically, there is no inherent ordering to the independent components. Also, the transform is applied independently to each recorded EEG snippet, making the subsequent analysis a harder problem. To mitigate this, an ordering can be fixed by sorting the independent components by how much variance of the original signal they explain (Variance Accounted For (VAF)).

2.4.3 Artifact classification and removal

Once a signal is decomposed into independent components, we gain a potent tool for artifact removal. By classifying a component into what type of activity it represents — e.g., eye blinks, heartbeats, muscle activity, brain activity — we can choose to remove the components that are classified as artifacts and reconstruct the signal without them. The classification can be done with a neural network trained on expert-labeled data[30]. Modern-day trained classifiers like this are a common inclusion in EEG processing libraries.

2.4.4 Normalization

There are a few reasonable ways to normalize the signal, such as z-score normalization, min-max normalization, or robust normalization. Because of the non-stationarity of the signal, it is common to apply normalization on a per-window basis, where a window is a short segment of the full recording. What remains to

be chosen is whether the normalization should apply per channel independently or jointly across all channels. Simply removing the mean across all channels is called rereferencing.

Chapter 3

Methods

3.1 EEG Preprocessing

First, we filter the EEG signal with a notch filter at the line frequency (50Hz). Then we apply a bandpass filter between 0.5Hz and 50Hz. We rereference the data. Then we apply the ICA transform, removing artifact components and reducing the dimensionality from 128 channels to 12. We calculate the band power for every ICA component and the standard EEG frequency bands, and concatenate the results. Without this step, we observe that the models struggle to train. Before concatenation, we apply z-score normalization per frequency band.

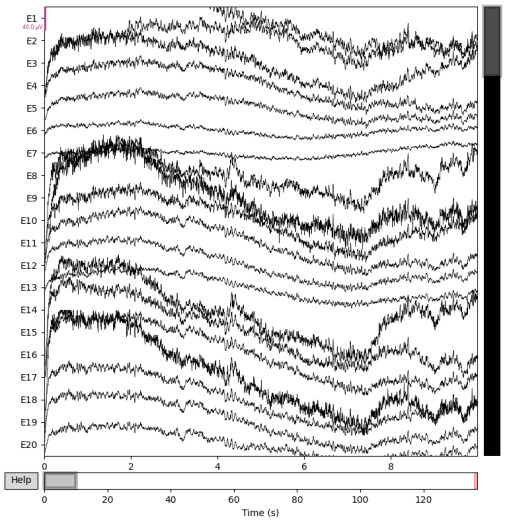

Figure 3.1: EEG signal: 10 second snippet of raw unprocessed EEG signal.

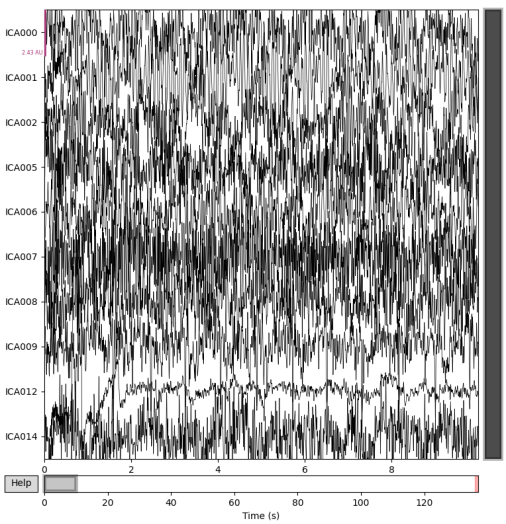

Figure 3.2: EEG signal: ICA components.

During training, we sample an EEG signal snippet of 3 seconds in length from the full recording. It was found that varying the length between 3s, 5s, 10s, and 20s

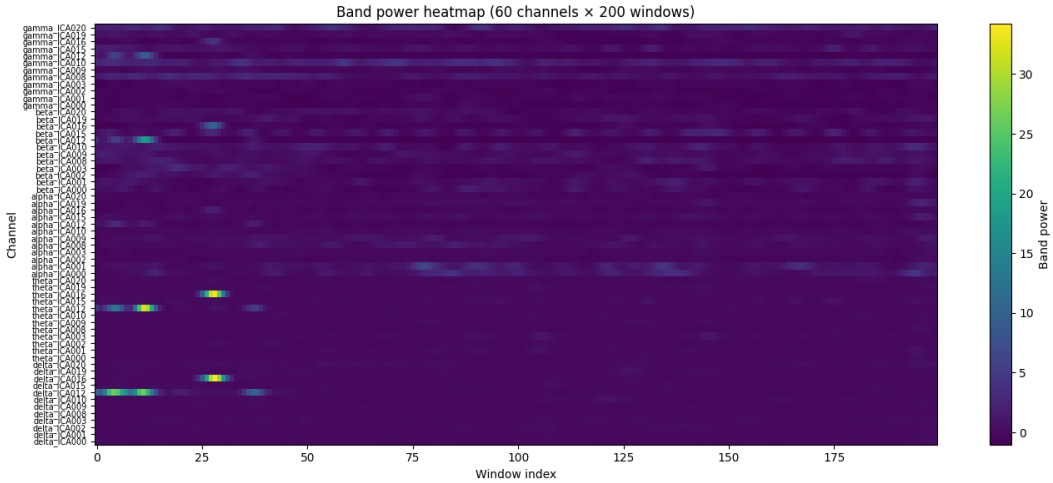

Figure 3.3: Band power of ICA components: 20 second snippet of preprocessed signal.

impacts classification performance, with longer snippets yielding better accuracy scores[29]. This acts like a data augmentation method. To avoid the Birthday Problem when randomly sampled segments often strongly overlap over part of the full recording, we use stratified sampling method, where first one of k segments is picked (one after another) and then a random snippet starting within chosen segment is sampled, spreading the distribution more evenly across the whole recording.

3.2 Note onset feature extraction

For the task of detecting note onsets, we use the essentia[3] library with its HFC onset detection method to transform the raw wave audio signal into a list of note onset start times.

3.3 Models

We use four different model types: KNN, SVM, XGBoost and a Convolutional Neural Network.

For a discussion of the KNN, SVM, and XGBoost models, we refer the reader to Sarker[36][35].

We define three related CNN networks: two variants for the classification task (taking either band-power processed data or raw EEG data as input) and one for the music reconstruction task. The classification networks are convolutional neural networks with 3 convolutional layers and a fully-connected layer. They differ in

the input dimensionality they are designed to work with: one takes EEG images after preprocessing, including band power calculation, which results in 1s snippets of shape 60×10. The other takes raw EEG data, which has a shape of 129×256 and is used for the ablation study in the appendix. The models have 35,188 and 64,660 parameters, respectively.

Layer Type	**Filter Size**	**Input → Output**
Conv2D (freq)	1×5	$[B, 1, 60, 10] \rightarrow [B, 16, 60, 10]$
BatchNorm2D	-	$''$
Conv2D (time)	5×1	$[B, 16, 60, 10] \rightarrow [B, 32, 60, 10]$
ELU	-	$''$
MaxPool2D	2×2	$[B, 32, 60, 10] \rightarrow [B, 32, 30, 5]$
Dropout	-	$''$
Conv2D (freq)	1×3	$[B, 32, 30, 5] \rightarrow [B, 64, 30, 5]$
BatchNorm2D	-	$''$
Conv2D (time)	3×1	$[B, 64, 30, 5] \rightarrow [B, 128, 30, 5]$
ELU	-	$''$
MaxPool2D	2×2	$[B, 128, 30, 5] \rightarrow [B, 128, 15, 2]$
Dropout	-	$''$
Conv2D (freq)	1×3	$[B, 128, 15, 2] \rightarrow [B, 128, 15, 2]$
BatchNorm2D	-	$''$
Conv2D (time)	3×1	$''$
ELU	-	$''$
MaxPool2D	2×2	$[B, 128, 15, 2] \rightarrow [B, 128, 7, 1]$
Dropout	-	$''$
Flatten	-	$[B, 128, 7, 1] \rightarrow [B, 896]$
Linear	-	$[B, 896] \rightarrow [B, 4]$

Table 3.1: CNNClassifier model: Classifier model trained on band power data.

The reconstruction network uses a U-Net style architecture with a 3-layer contracting encoder, a middle bottleneck fully connected layer, and a 3-layer expanding decoder. The encoder has the same architecture as the classifier model, and the decoder is correspondingly a 3-layer upsampling convolutional network. There are skip connections between matching layers of the encoder and decoder. Because the input has a different dimensionality than the output and consequently, the outputs of the encoder and decoder layers differ in shape, the skip connections also apply interpolation to match the dimensions.

Layer Type	**Filter Size**	**Input → Output**
		Encoder
Conv2D (freq)	1×5	$[B, 1, 60, 10] \rightarrow [B, 16, 60, 10]$
BatchNorm2D	-	$''$
Conv2D (time)	5×1	$[B, 16, 60, 10] \rightarrow [B, 32, 60, 10]$
ELU	-	$''$
MaxPool2D	2×2	$[B, 32, 60, 10] \rightarrow [B, 32, 30, 5]$
Dropout	-	$''$
Conv2D (freq)	1×3	$[B, 32, 30, 5] \rightarrow [B, 64, 30, 5]$
BatchNorm2D	-	$''$
Conv2D (time)	3×1	$[B, 64, 30, 5] \rightarrow [B, 128, 30, 5]$
ELU	-	$''$
MaxPool2D	2×2	$[B, 128, 30, 5] \rightarrow [B, 128, 15, 2]$
Dropout	-	$''$
Conv2D (freq)	1×3	$[B, 128, 15, 2] \rightarrow [B, 128, 15, 2]$
BatchNorm2D	-	$''$
Conv2D (time)	3×1	$''$
ELU	-	$''$
MaxPool2D	2×2	$[B, 128, 15, 2] \rightarrow [B, 128, 7, 1]$
Dropout	-	$''$
		Bottleneck
Flatten	-	$[B, 128, 7, 1] \rightarrow [B, 896]$
Linear + ELU	-	$[B, 896] \rightarrow [B, 128]$
Unsqueeze	-	$[B, 128] \rightarrow [B, 128, 1, 1]$
ConvTranspose2D	7×1	$[B, 128, 1, 1] \rightarrow [B, 128, 7, 1]$
		Decoder
Concat (skip enc3)	-	$[B, 128, 7, 1] + [B, 128, 7, 1] \rightarrow [B, 256, 7, 1]$
Upsample	2×4	$[B, 256, 7, 1] \rightarrow [B, 256, 14, 4]$
Conv2D (time)	3×1	$[B, 256, 14, 4] \rightarrow [B, 128, 14, 4]$
BatchNorm2D	-	$''$
Conv2D (freq)	1×3	$''$
ELU	-	$''$
Dropout	-	$''$
AdaptiveAvgPool2D	-	$[B, 128, 14, 4] \rightarrow [B, 128, 15, 4]$
Concat (skip enc2)	-	$[B, 128, 15, 4] + [B, 128, 15, 4] \rightarrow [B, 256, 15, 4]$
Upsample	2×4	$[B, 256, 15, 4] \rightarrow [B, 256, 30, 16]$
Conv2D (time)	3×1	$[B, 256, 30, 16] \rightarrow [B, 64, 30, 16]$
BatchNorm2D	-	$''$
Conv2D (freq)	1×3	$[B, 64, 30, 16] \rightarrow [B, 32, 30, 16]$
ELU	-	$''$
Dropout	-	$''$
Concat (skip enc1)	-	$[B, 32, 30, 16] + [B, 32, 30, 16] \rightarrow [B, 64, 30, 16]$
Upsample	2×4	$[B, 64, 30, 16] \rightarrow [B, 64, 60, 64]$
Conv2D (time)	5×1	$[B, 64, 60, 64] \rightarrow [B, 16, 60, 64]$
BatchNorm2D	-	$''$
Conv2D (freq)	1×5	$[B, 16, 60, 64] \rightarrow [B, 1, 60, 64]$
ELU	-	$''$
Dropout	-	$''$
		Final Adaptation
Conv2D + ELU	3×3	$[B, 1, 60, 64] \rightarrow [B, 1, 60, 64]$
AdaptiveAvgPool2D	-	$[B, 1, 60, 64] \rightarrow [B, 1, 64, 64]$
Concat (skip input)	-	$[B, 1, 64, 64] + [B, 1, 64, 64] \rightarrow [B, 2, 64, 64]$
Conv2D	1×1	$[B, 2, 64, 64] \rightarrow [B, 1, 64, 64]$

Table 3.2: CNNReconstruction model: U-Net style network for music spectrogram reconstruction.

Chapter 4

Datasets

We test our hypotheses on two datasets: MUSIN-G[26] and what we call BCMI-training[7].

4.1 MUSIN-G

It is a dataset of joint music and EEG recordings of 20 non-musician subjects, aged 22-28 years. Each subject was presented with 12 songs of various genres, ranging from Indian Folk to Electronic Dance.

4.2 BCMI-training

BCMI-training is a dataset of joint music and EEG recordings of 10 participants aged 19-30. It is part of a series of datasets collected during studies investigating human reactions to affective musical stimuli and whether reported emotionality can be influenced by music[8]. The music played in this study is fully computer-generated, and no track repeats across the dataset. The musical snippets are tagged with emotionality labels (9 possibilities on valence/arousal grid) which are inputs to the music generation algorithm. In the study, subjects listened to a musical snippet of about 20 seconds in length and immediately afterward to another snippet of different emotionality. For our purposes, we split such trials into two separate recordings, one for each snippet.

4.3 Dataset processing

In addition to the methods, we describe how a dataset is loaded for a study using our library.

We unify the various analyzed datasets under the ‘EEGMusicDataset‘ class, which provides a common interface to access the data and metadata of the recordings. Then, various preprocessing functions are applied to the data; their evaluation is lazy. Lastly, the ‘StratifiedSamplingDataset‘ wrapper is applied, which samples the data in a stratified way, as described in the methods chapter.

```
ds = EEGMusicDataset.load_ondisk(Path("..."))

ds = MappedDataset(ds, lambda x:
  ica_band_power_trial(
    rereference_trial(
      filter_trial(
        notch_filter_trial(x, 50.0), l_freq=0.5, h_freq=50.0
      )
    ),
    n_components=12,
    bands = [(0.5, 4), (4, 8), (8, 13), (13, 30), (30, 45)],
    window_sec= 2.0,
    hop_sec = 0.1,
  )
)

ds = StratifiedSamplingDataset(ds, n_strata=10, trial_length_secs=Fraction(3, 1))
```

Chapter 5

Related work

5.1 Music classification

Classification of listened music based on EEG signal has been tackled in various forms by several researchers.

As early as 2011, a linear logistic-regression classifier was successfully used to classify ERP (event-related potential) segments of EEG signals into listened-to songs[37]. The focus on ERPs means that every trial, which is slightly over 3 seconds long, acts as its own snippet for classification. Therefore, the EEG snippets are aligned in time — all starting when the music began playing in the experimental scenario.

In a similar paradigm, Sterni et al.[39] classify listened-to music using first a dimensionality reduction technique to learn signal components, and then a CNN with a single convolutional layer. They also succeed in classifying trials based on the meter of a song (3/4 versus 4/4).

Two studies, linked by the collected dataset later published as MUSIN-G, tackle the problem of classifying listened-to music based on the EEG signal in a paradigm similar to ours, where a snippet for classification is randomly sampled from the entire recording. A study by Lawhatre uses a multilevel wavelet method, which finds frequency components at various time scales, to extract lower-level features and then tests various models such as KNN, decision trees, SVM, and CNNs[21]. The study establishes the usefulness of the dataset.

The second study applies a 3-layer CNN to either the time-domain EEG signal or the frequency-domain spectrogram of the EEG[38]. It does not perform better than a random classifier in the time-domain case. However, in the frequency-domain case, the best CNN model with 1.6M parameters achieved over 80% accuracy. In the cross-participant setting, however, the model failed to generalize.

The existence of a signal related to music in the EEG signal poses a question:

what about imagining music? In a 2021 paper, researchers attempted to classify EEG by first decoding EEG from music[22]. The musical input is transformed into a binary vector that encodes note onsets, together with an auxiliary vector that encodes expected note surprise, as calculated by a mathematical model of musical expectation in sheet music. For every music track, the predicted matching EEG is compared against the real EEG signal, and the one with the strongest match is chosen. Separate models were trained for the listened and imagined music, but it was also shown that they relate to each other, with the listened version being a lagged version of the imagined one.

5.2 Representation learning

Stober et al. approach a more general problem of learning representations of the EEG signal with a convolutional autoencoder[40]. To combat the small and noisy dataset, the model is trained on a given trial to reconstruct a possibly different trial, but of the same subject and stimulus. They are able to train CNN and SVC classifiers to predict the song ID from the hidden representation of the autoencoder.

This raises the question of whether a common representation space could be found for both music and the EEG signal, or if the two can be related to each other. Researchers attempted to answer the second question by comparing the distance matrices within the sets of extracted audio features and extracted EEG features[12]. The study reports mixed results.

5.3 Music reconstruction

Works of the last few years tackled the harder problem of reconstructing listened music from the EEG signal.

The simplest approach seems to be directly training a function that maps the EEG signal to music. A 2022 study attempts this with a 5-layer CNN model[32]. The network is fed 1s snippets, randomly sampled from the full recording and assigned to the training set with a 75% probability, and outputs audio in the form of a mel spectrogram. The accuracy of the model is assessed by training a classifier to predict the song ID from the reconstructed music, achieving over 80% accuracy. The model outputs were also evaluated by human listeners, who were able to successfully determine which of two presented audio clips corresponded to a third target sound.

The diffusion approach to music generation was tested in a 2025 study[31]. The music is generated via a diffusion process in the latent space, where initial noise is iteratively refined with a denoising network. The denoised latent variable is then fed into a decoder that converts it into an audio spectrogram. The denoising model is a pretrained AudioLDM2 model, trained on a general music generation task. The

researchers trained a ControlNet[46] adapter to condition the denoising network on the EEG signal. The results are quantified using metrics applicable to generation tasks, such as FAD or the Pearson correlation coefficient.

There are also attempts at combining EEG with fMRI data[6]. The fMRI signal allows for the localization of sources of neural activity. EEG processing can use this information to translate the EEG signal into an equivalent one, as if it were recorded with electrodes at the sites of activity. This processed EEG constitutes the input to a biLSTM model. The model is able to reconstruct the music in waveform. A separate model is trained for every study subject. The output quality is evaluated by calculating the similarity between the reconstructed audio and all the listened-to songs, and then ranking them based on similarity. Importantly, even without the fMRI-influenced processing step, the results are positive, albeit worse.

Only fMRI was used in a 2023 study[9], which relies on the MuLan music encoder and the MusicLM music decoder. Here, the chosen voxels of the fMRI signal are an input to a linear regression model that is trained to predict the embeddings of the music as given by the MuLan encoder model. Then, the MusicLM model generates audio from the predicted embeddings.

Described attempts show that the task while hard is not impossible.

Chapter 6

Results

6.1 Song name prediction

We start with the task of predicting the song name of a listened-to song based on the EEG signal. We split the MUSIN-G dataset into recordings of 12 subjects for training and the remaining 8 subjects for testing (for CNN, half of the test set serves as the validation set). We then train a model to predict the song name from the EEG signal. We achieve the below results for each variant of the tested model.

Model	**Accuracy**
KNN	0.1374
SVM	0.1229
XGBoost	0.1668
CNN	0.1740

Table 6.1: Song name prediction accuracy: subject-wise split.

Our results are comparable to the study by Pandey et al.[29], in which researchers achieved 12.5% mean subject accuracy (a random classifier gives $\frac{1}{12} = 0.08(3)$).

In the above study, researchers also tested scores achieved in a different testing paradigm. They split the MUSIN-G dataset into training and testing by taking the initial n seconds of each recording as training data and the remaining seconds as testing data. The study compares results that can be achieved for a few chosen n parameters $(3, 5, 10, 20)$, achieving higher accuracy for larger training partitions. The results in this variant of the experiment are much higher (84%) than in the previous variant. We are able to reproduce these results, and for $n = 20$ we achieve accuracies as shown in the table:

This poor generalization suggests that the model in this second variant does not pick up on neural correlates of the music but rather on some trial-specific aspects of

Model	Accuracy
XGBoost	0.85
SVM	0.57
KNN	0.64
NN	-

Table 6.2: Song id prediction accuracy in the MUSIN-G dataset: temporal split.

the EEG recording. We test the hypothesis with two experiments.

First, we run a similar experiment on the BCMI-training dataset. The dataset contains from 144 up to 432 trials recorded for a single subject. The recordings share the "emotionality" of the listened-to music, but no listened-to song repeats across the dataset as the songs are computer-generated for each trial. We find that, similarly to the study, classifiers are able to predict the trial ID from the EEG signal with high accuracy.

Model	Accuracy
XGBoost	0.78
SVM	0.91
KNN	0.88
NN	—

Table 6.3: Song id prediction experiment in the single subject variant in the BCMI-training dataset.

6.1.1 DTW-based trial id prediction

Secondly, guided by the intuition that the classifiers notice directly repeatable patterns in the EEG signal between training and test sets and are not forced to generalize beyond that, we propose the following experiment. We attempt to predict the trial ID using the DTW algorithm.

DTW in its most basic form is a method of measuring the distance between two time series by finding an optimal alignment between them. Such an alignment is a match between points of one time series and points of the other time series that minimizes the sum of distances between matched points, starts and ends at the two matched ends of the two time series, and is monotonic in between. We employ a generalized version called approximate-subsequence-DTW, which can be defined by:

1. Given two sequences A and B, it finds a match between an infix of A and B.
2. The summed distances between the matched points are within the cost allowance.
3. The matched infix is of maximal length.

For a single subject of the BCMI-training dataset, we predict the trial ID of a test trial by finding, across all training trials, the one that scores the highest when calculating the approximate-subsequence-DTW distance between the test trial and the training trial. We fit the cost allowance hyperparameter on the validation set and then test the accuracy of the method on the test set. With this simple method we achieve 50.83% test accuracy. The experiment is meant to showcase that such direct pattern matching is possible and therefore—it is possible for the model to learn it and rely on it greatly, with little generalization capability.

6.2 Emotion prediction

The BCMI-training dataset is made of computer-generated snippets of music. These musical snippets are tagged with emotionality labels, one of 9 possibilities arranged on Valence and Arousal axes. To compare the difficulty of this task with previous tasks, we can think of emotionality labels as abstraction classes that join multiple individual trials together. Therefore, it is easier to predict an emotion label than an identifier, as this requires less precision, but during training, the reward per training step is less valuable and gives less information. The balance does not seem to be in favor of this task, as we are unable to achieve good results, even though we do in the prediction task.

Method	Accuracy
XGBoost	0.1089
SVM	0.1123
KNN	0.1117
CNN	0.1130

Table 6.4: Emotion prediction accuracy

6.3 Music "density" prediction

Looking for musical features visible through the EEG recording, we define a "density" of a music snippet and, by extension, the paired EEG snippet. First, we mark note onsets in the music recordings. We do that by looking at the music's mel spectrograms and marking the time points at which the spectrogram's energy increases significantly, weighting the changes in the high frequencies more (High-Frequency-Content method). The HFC method was visually compared to other methods and chosen as seemingly best.

Now, for a chosen length of a time window, we get a certain distribution of the number of note onsets within windows of this length. Calculating the median number, we can split the snippets into two classes: the "sparse" class with a number

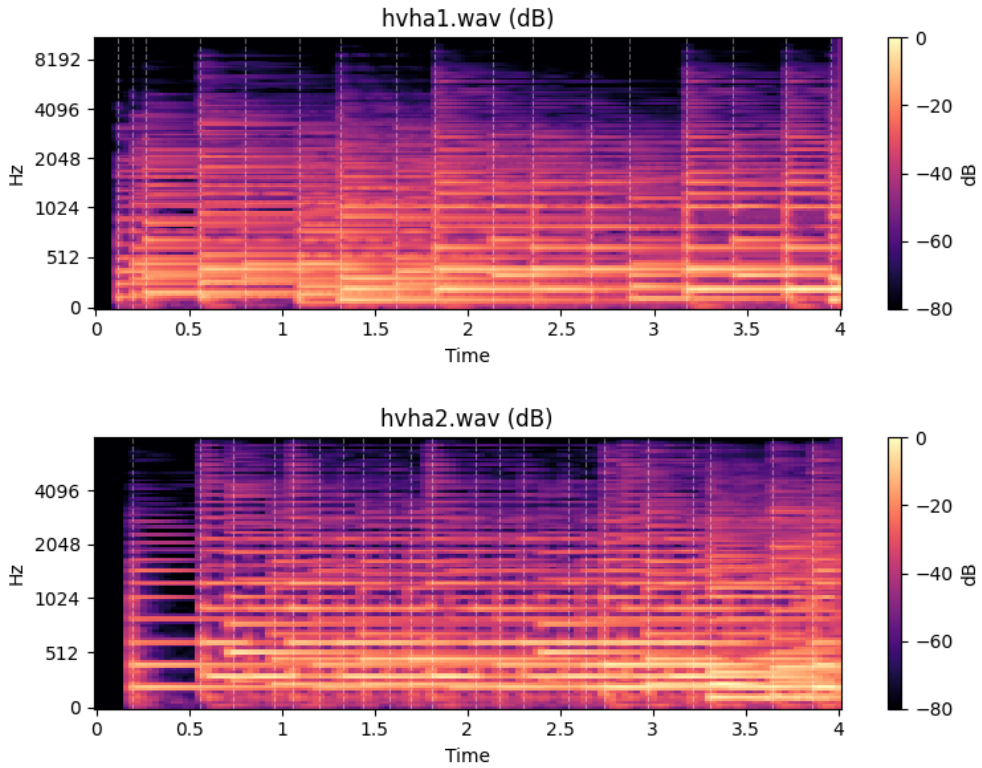

Figure 6.1: Spectrograms of two 4s music snippets with marked note onsets.

of onsets below the median, and the "dense" class with a number of onsets above the median. We calculate the median only on the training set and define a classification task of predicting the class of a snippet based on the paired EEG snippet. We hope that success in this task may be a stronger indicator that some musical features are present and can be decoded from the EEG signal.

Results are shown in the table below.

Model	**Accuracy**
XGBoost	0.5407
SVM	0.5093
KNN	0.5264
CNN	0.5206

Table 6.5: Music density prediction accuracy.

6.4 Music reconstruction

We attempt to reconstruct listened-to music from the EEG signal for the MUSIN-G dataset. We treat this task as an extension of the song ID prediction task: if a

model can learn to predict a song, can it learn to predict the specific fragment of a song and generate its mel spectrogram? Our architecture of the $CNN_{reconstruction}$ model used for this task reflects this idea. It can be seen as an encoder and decoder model, with skip connections, where the encoder is similar to the $CNN_{classification}$ model.

Reflecting the results in the song identification tasks, we observe that the model is able to reconstruct music to some extent in the variant where single-subject recordings are split into training and test partitions by assigning the initial 20s of each recording to training. The reconstructions are accurate enough to allow for meaningful classification (see figures below).

In the cross-subject variant, where we investigate the ability for cross-subject generalization and a different portion of subjects is used for training and testing, we don't achieve any meaningful results in the reconstruction task. This matches the established accuracies in the song ID prediction task for this variant, which were much lower and were all between 12% and 18%, compared to the 57% to 85% range.

Accuracy scores in the temporal-split

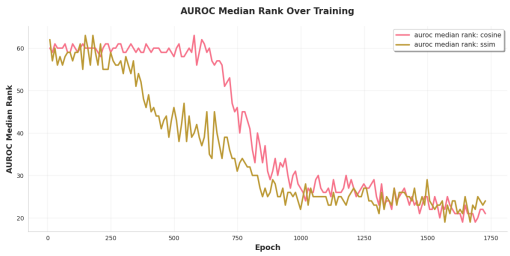

Figure 6.2: For a generated mel spectrogram, we calculate similarities with all 120 real spectrograms from the current batch. We then sort the spectrograms based on the similarity scores and find out what the rank of the true spectrogram is to assess the quality of our predicted one. The plot shows the median rank for two used similarity measures: cosine similarity and structural similarity.

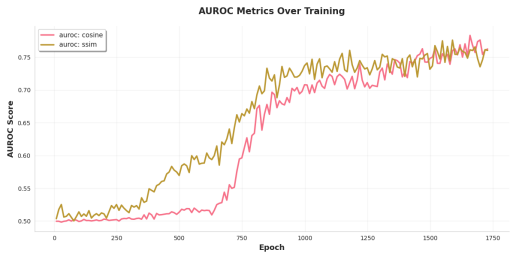

Figure 6.3: AUROC score counts the ratio of times when the real spectrogram is ranked in the first half of the sorted order. It can be interpreted as the probability that the predicted spectrogram is closer to the real spectrogram than to a randomly sampled spectrogram.

Looking at the reconstructed spectrograms, we can notice that the network has not yet learned the high-resolution temporal aspects of the songs. It cannot mark note onsets accurately. Instead, it tends to produce smooth spectrograms with almost equally colored rows that show the average power in a frequency band.

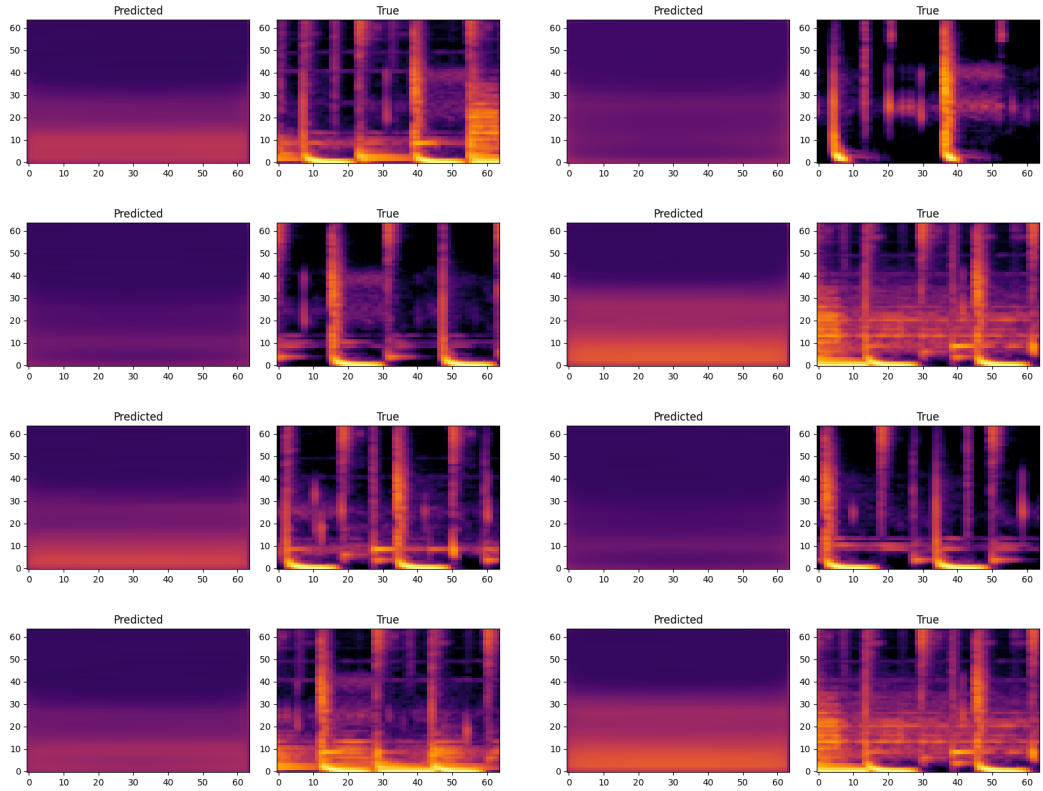

Figure 6.4: Reconstructed spectrograms.

Chapter 7

Discussion

We have shown the capability of machine learning models to extract music features from EEG data. We have succeeded in the tasks of song identification, song reconstruction, and a proposed task of music "density" prediction.

Nevertheless, the capability seems limited. Song identification and reconstruction struggle to generalize across subjects, and song reconstruction lacks high-resolution temporal details. The success of the DTW experiment casts doubt on the claim that the song identification task requires extraction of higher-level features. We propose a task of music "density" prediction to test the ability for purely musical feature prediction. We achieve success in that task, but the seemingly mediocre performance of the density prediction model doesn't make the claim fully convincing.

Appendix A

Ablation studies

A.1 Stratified Sampling

When sampling snippets of specified length from the dataset, we use a stratified sampling approach. We do this to avoid draws where a snippet is repeatedly sampled from a single region of a trial. Such an imbalance could lead to overfitting to a portion of the dataset and, generally, to treating various regions disproportionately. We study the effect of this approach and find that it may matter less than we anticipated.

Recall that the stratified approach means sampling in n_{strata} consecutive turns from n_{strata} disjoint regions of the sampled dataset. The parameter n_{strata} controls the granularity or the number of disjoint regions summing up to the full trial length. We perform an experiment where we always sample 100 times from each trial, but vary the number of strata n_{strata}. We test on the task of song identification in the MUSIN-G dataset. The results are shown in the graph below and, at most, seem to suggest a detrimental effect of such sampling when used with the XGBoost model. The effect seems small enough that we can follow the intuition behind the stratified approach and use it regardless.

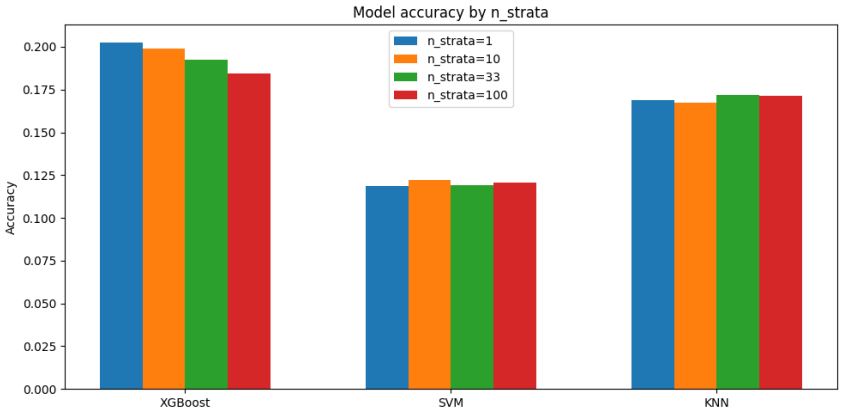

Figure A.1: Effect of stratified sampling on classification accuracy.

A.2 ICA

In the preprocessing of the EEG data, we use ICA as a dimensionality reduction transform on the EEG signal. ICA finds a linear transform of the EEG signal into so-called independent components, which are ordered by importance. Similar to the use of PCA, we can drop the bottom components and reduce the number of channels in the EEG data. We compare this use of ICA in preprocessing to picking a set of channels and dropping the rest. The set of channels to be picked in this "No ICA" case is predefined and chosen based on proximity to auditory processing regions of the brain or a known relation to music processing.

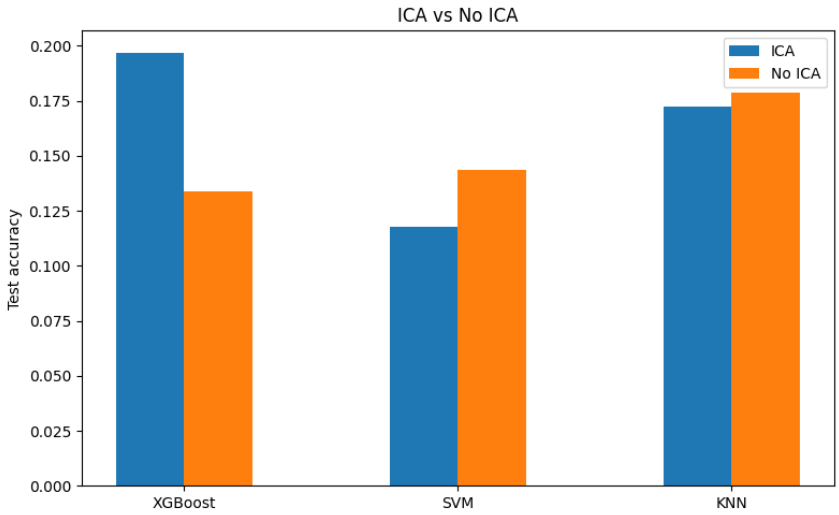

Figure A.2: Effect of ICA on classification accuracy.

The results as tested on the task of song identification in the MUSIN-G dataset are inconclusive.

Bibliography

[1] Eugene Aserinsky and Nathaniel Kleitman. Regularly occurring periods of eye motility, and concomitant phenomena, during sleep. 1953. *The Journal of neuropsychiatry and clinical neurosciences*, 15(4):454–455, Fall 2003.

[2] Erol Basar and Bahar Güntekin. *Review of delta, theta, alpha, beta, and gamma response oscillations in neuropsychiatric disorders.* 01 2013.

[3] Dmitry Bogdanov, Nicolas Wack, Emilia Gómez, Sankalp Gulati, Perfecto Herrera, Oscar Mayor, Gerard Roma, Justin Salamon, José Zapata, and Xavier Serra. Essentia: an open-source library for sound and music analysis. In *Proceedings of the 21st ACM International Conference on Multimedia*, MM '13, page 855–858, New York, NY, USA, 2013. Association for Computing Machinery.

[4] Richard Caton. Electrical currents of the brain. *The Chicago Journal of Nervous & Mental Disease*, 2(4):610, October 1875.

[5] Alastair Compston. The berger rhythm: potential changes from the occipital lobes in man, by e.d. adrian and b.h.c. matthews (from the physiological laboratory, cambridge). brain 1934: 57; 355-385. *Brain*, 133:3–6, 01 2010.

[6] Ian Daly. Neural decoding of music from the eeg. *Scientific reports*, 13(1):624, Jan 2023.

[7] Ian Daly, Nicoletta Nicolaou, Duncan Williams, Faustina Hwang, Alexis Kirke, Eduardo Miranda, and Slawomir J. Nasuto. "a dataset recorded during development of an affective brain-computer music interface: training sessions", 2020.

[8] Ian Daly, Nicoletta Nicolaou, Duncan Williams, Faustina Hwang, Alexis Kirke, Eduardo Miranda, and Slawomir J. Nasuto. Neural and physiological data from participants listening to affective music. *Scientific Data*, 7(1):177, Jun 2020.

[9] Timo I. Denk, Yu Takagi, Takuya Matsuyama, Andrea Agostinelli, Tomoya Nakai, Christian Frank, and Shinji Nishimoto. Text-to-music generation models capture musical semantic representations in the human brain. *Nature Communications*, 17(1), November 2025.

[10] Andreas K. Engel and Pascal Fries. Beta-band oscillations–signalling the status quo? *Current opinion in neurobiology*, 20(2):156–165, Apr 2010.

[11] Robert S. Fisher, J. Helen Cross, Carol D'Souza, Jacqueline A. French, Sheryl R. Haut, Norimichi Higurashi, Edouard Hirsch, Floor E. Jansen, Lieven Lagae, Solomon L. Moshé, Jukka Peltola, Eliane Roulet Perez, Ingrid E. Scheffer, Andreas Schulze-Bonhage, Ernest Somerville, Michael Sperling, Elza Márcia Yacubian, and Sameer M. Zuberi. Instruction manual for the ilae 2017 operational classification of seizure types. *Epilepsia*, 58(4):531–542, Apr 2017.

[12] Chris Foster, Dhanush Dharmaretnam, Haoyan Xu, Alona Fyshe, and George Tzanetakis. Decoding music in the human brain using eeg data. In *2018 IEEE 20th International Workshop on Multimedia Signal Processing (MMSP)*, pages 1–6, 2018.

[13] Pascal Fries. Fries, p.: A mechanism for cognitive dynamics: Neuronal communication through neuronal coherence. trends in cognitive sciences 9, 474-480. *Trends in cognitive sciences*, 9:474–80, 11 2005.

[14] P. Gloor. Long-term monitoring in epilepsy. *Electroencephalography and clinical neurophysiology. Supplement*, 37:xiii–xx, 1985.

[15] Thalia Harmony. The functional significance of delta oscillations in cognitive processing. *Frontiers in Integrative Neuroscience*, Volume 7 - 2013, 2013.

[16] Sharon Keenan and Max Hirshkowitz. Chapter 141 - monitoring and staging human sleep. In Meir H. Kryger, Thomas Roth, and William C. Dement, editors, *Principles and Practice of Sleep Medicine (Fifth Edition)*, pages 1602–1609. W.B. Saunders, Philadelphia, fifth edition edition, 2011.

[17] Wolfgang Klimesch. EEG alpha and theta oscillations reflect cognitive and memory performance: a review and analysis. *Brain Research Reviews*, 29(2–3):169–195, April 1999.

[18] Meir H. Kryger, T. Roth, and William C. Dement. *Principles and Practice of Sleep Medicine*. Saunders, Philadelphia, 3rd edition, 2000.

[19] Jim Lagopoulos, Jian Xu, Inge Rasmussen, Alexandra Vik, Gin S Malhi, Carl F Eliassen, Ingrid E Arntsen, Jardar G Saether, Stig Hollup, Are Holen, Svend Davanger, and Øyvind Ellingsen. Increased theta and alpha EEG activity during nondirective meditation. *Journal of Alternative and Complementary Medicine*, 15(11):1187–1192, November 2009.

[20] Saroj Lal and Ashley Craig. Electroencephalography activity associated with driver fatigue: Implications for a fatigue countermeasure device. *Journal of Psychophysiology*, 15:183–189, 07 2001.

[21] Prashant Lawhatre, Bharatesh R Shiraguppi, Esha Sharma, Krishna Prasad Miyapuram, and Derek Lomas. Classifying songs with eeg, 2020.

[22] Guilhem Marion, Giovanni M. Di Liberto, and Shihab A. Shamma. The music of silence: Part i: Responses to musical imagery encode melodic expectations and acoustics. *Journal of Neuroscience*, 41(35):7435–7448, 2021.

[23] Carlo Matteucci. Xi. electro-physiological researches. —first memoir. the muscular current. *Philosophical Transactions of the Royal Society of London*, (135):283–295, 12 1845.

[24] Carlo Matteucci. Xxxi. electro-physiological resarches. on induced contraction.-ninth series. *Philosophical Transactions of the Royal Society of London*, (140):645–649, 12 1850.

[25] J. A. McEwen and G. B. Anderson. Modeling the stationarity and gaussianity of spontaneous electroencephalographic activity. *IEEE transactions on biomedical engineering*, 22(5):361–369, Sep 1975.

[26] Krishna Prasad Miyapuram, Nashra Ahmad, Pankaj Pandey, and James Derek Lomas. Electroencephalography (eeg) dataset during naturalistic music listening comprising different genres with familiarity and enjoyment ratings. *Data in Brief*, 45:108663, 2022.

[27] Faisal Mushtaq, Dominik Welke, Anne Gallagher, Yuri G. Pavlov, Layla Kouara, Jorge Bosch-Bayard, Jasper J. F. van den Bosch, Mahnaz Arvaneh, Amy R. Bland, Maximilien Chaumon, Cornelius Borck, Xun He, Steven J. Luck, Maro G. Machizawa, Cyril Pernet, Aina Puce, Sidney J. Segalowitz, Christine Rogers, Muhammad Awais, Claudio Babiloni, Neil W. Bailey, Sylvain Baillet, Robert C. A. Bendall, Daniel Brady, Maria L. Bringas-Vega, Niko A. Busch, Ana Calzada-Reyes, Armand Chatard, Peter E. Clayson, Michael X. Cohen, Jonathan Cole, Martin Constant, Alexandra Corneyllie, Damien Coyle, Damian Cruse, Ioannis Delis, Arnaud Delorme, Damien Fair, Tiago H. Falk, Matthias Gamer, Giorgio Ganis, Kilian Gloy, Samantha Gregory, Cameron D. Hassall, Katherine E. Hiley, Richard B. Ivry, Karim Jerbi, Michael Jenkins, Jakob Kaiser, Andreas Keil, Robert T. Knight, Silvia Kochen, Boris Kotchoubey, Olave E. Krigolson, Nicolas Langer, Heinrich R. Liesefeld, Sarah Lippé, Raquel E. London, Annmarie MacNamara, Scott Makeig, Welber Marinovic, Eduardo Martínez-Montes, Aleya A. Marzuki, Ryan K. Mathew, Christoph Michel, José D. R. Millán, Mark Mon-Williams, Lilia Morales-Chacón, Richard Naar, Gustav Nilsonne, Guiomar Niso, Erika Nyhus, Robert Oostenveld, Katharina Paul, Walter Paulus, Daniela M. Pfabigan, Gilles Pourtois, Stefan Rampp, Manuel Rausch, Kay Robbins, Paolo M. Rossini, Manuela Ruzzoli, Barbara Schmidt, Magdalena Senderecka, Narayanan Srinivasan, Yannik Stegmann, Paul M. Thompson, Mitchell Valdes-Sosa, Melle J. W. van der Molen, Domenica Veniero, Edelyn Verona, Bradley Voytek, Dezhong Yao, Alan C. Evans, and Pedro Valdes-Sosa. One hundred years of eeg for brain and behaviour research. *Nature human behaviour*, 8(8):1437–1443, Aug 2024.

[28] Sylvie Nozaradan, Younes Zerouali, Isabelle Peretz, and André Mouraux. Capturing with eeg the neural entrainment and coupling underlying sensorimotor synchronization to the beat. *Cerebral cortex (New York, N.Y. : 1991)*, 25, 10 2013.

[29] Pankaj Pandey, Gulshan Sharma, Krishna Miyapuram, Ram Subamanian, and Derek Lomas. Music identification using brain responses to initial snippets. pages 1246–1250, 05 2022.

[30] Luca Pion-Tonachini, Ken Kreutz-Delgado, and Scott Makeig. Iclabel: An automated electroencephalographic independent component classifier, dataset, and website. *NeuroImage*, 198:181–197, Sep 2019.

[31] Emilian Postolache, Natalia Polouliakh, Hiroaki Kitano, Akima Connelly, Emanuele Rodolà, Luca Cosmo, and Taketo Akama. Naturalistic music decoding from eeg data via latent diffusion models, 2025.

[32] Adolfo G. Ramirez-Aristizabal and Chris Kello. Eeg2mel: Reconstructing sound from brain responses to music, 2022.

[33] W. J. Ray and H. W. Cole. Eeg alpha activity reflects attentional demands, and beta activity reflects emotional and cognitive processes. *Science (New York, N.Y.)*, 228(4700):750–752, May 1985.

[34] Moises Rivera-Ruiz, Christian Cajavilca, and Joseph Varon. Einthoven's string galvanometer: the first electrocardiograph. *Texas Heart Institute journal*, 35(2):174–178, 2008.

[35] Iqbal H. Sarker. Deep learning: A comprehensive overview on techniques, taxonomy, applications and research directions. *SN computer science*, 2(6):420, 2021.

[36] Iqbal H. Sarker. Machine learning: Algorithms, real-world applications and research directions. *SN computer science*, 2(3):160, 2021.

[37] Rebecca S. Schaefer, Jason Farquhar, Yvonne Blokland, Makiko Sadakata, and Peter Desain. Name that tune: Decoding music from the listening brain. *NeuroImage*, 56(2):843–849, 2011. Multivariate Decoding and Brain Reading.

[38] Dhananjay Sonawane, Krishna Prasad Miyapuram, Bharatesh RS, and Derek J. Lomas. Guessthemusic: Song identification from electroencephalography response, 2020.

[39] Avital Sternin. Classifying music perception and imagination using eeg. 2016.

[40] Sebastian Stober, Avital Sternin, Adrian M. Owen, and Jessica A. Grahn. Deep feature learning for eeg recordings, 2016.

[41] C. Tallon-Baudry and O. Bertrand. Oscillatory gamma activity in humans and its role in object representation. *Trends in cognitive sciences*, 3(4):151–162, Apr 1999.

[42] Carlo Alberto Tassinari. An electroencephalographer recalls the history of the federation on the 70th anniversary of its journal, clinical neurophysiology. *Clinical Neurophysiology*, 130(12):2258–2263, 2019.

[43] Mario Tudor, Lorainne Tudor, and Katarina Ivana Tudor. [hans berger (1873-1941)–the history of electroencephalography]. *Acta medica Croatica : casopis Hravatske akademije medicinskih znanosti*, 59(4):307–313, 2005.

[44] Prasad Vannemreddy, James L. Stone, and Konstantin V. Slavin. Frederic gibbs and his contributions to epilepsy surgery and electroencephalography. *Neurosurgery*, 70(3):774–782, Mar 2012.

[45] J. J. Vidal. Toward direct brain-computer communication. *Annual Review of Biophysics and Bioengineering*, 2:157–180, 1973.

[46] Lvmin Zhang, Anyi Rao, and Maneesh Agrawala. Adding conditional control to text-to-image diffusion models, 2023.